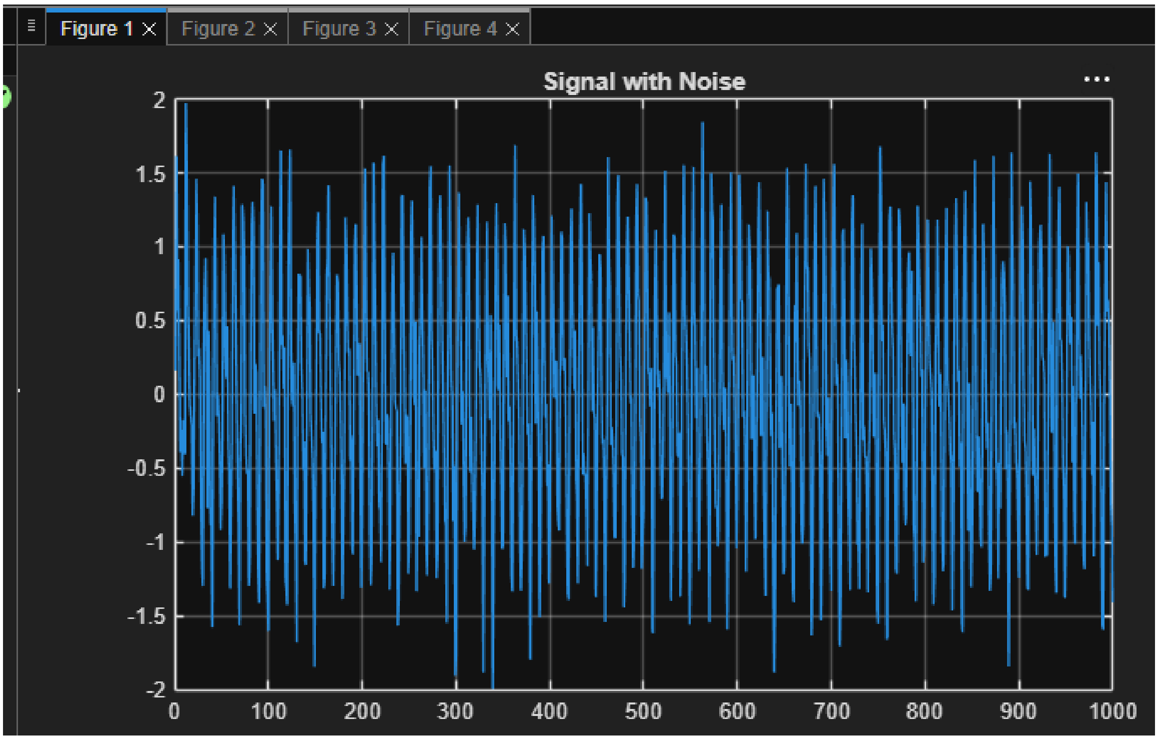
Experiment 7.1: Power Spectral Density Estimation Using Periodogram in MATLAB

Aim: To estimate the power spectral density of a noisy signal using the periodogram method and analyze its spectrum.

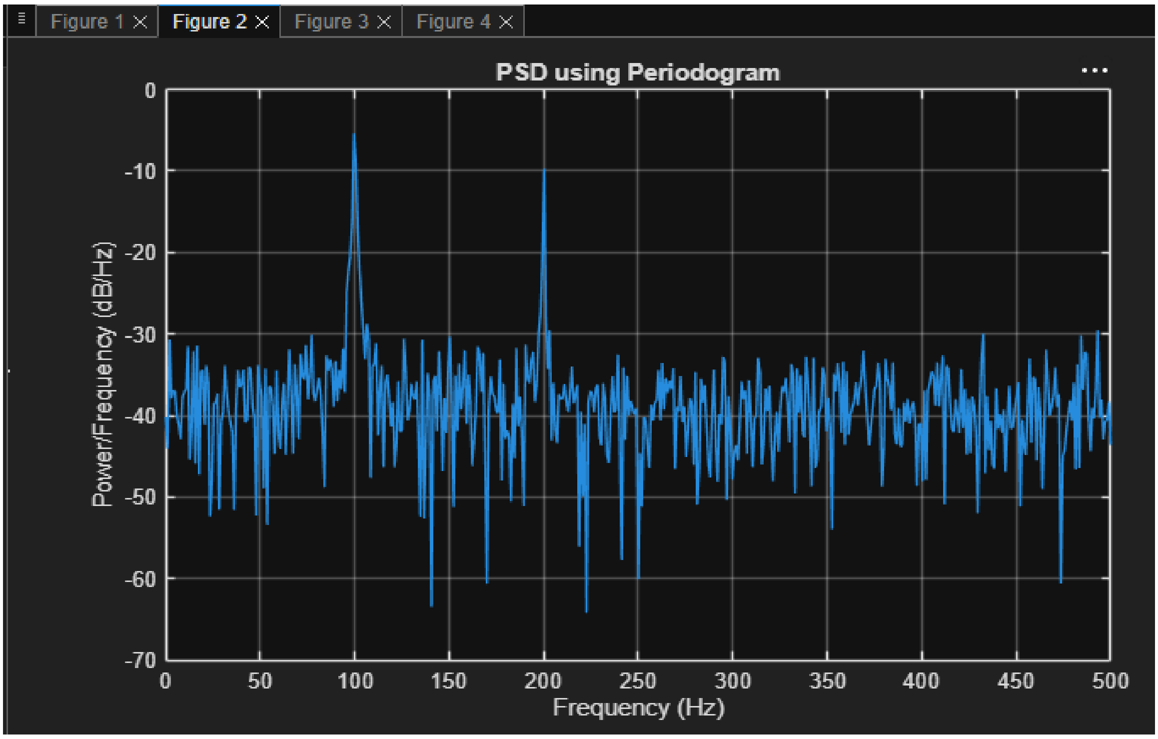
MATLAB Code

```
clc; clear; close all
fs=1000; t=(0:999)/fs;
s=sin(2*pi*100*t)+0.5*sin(2*pi*200*t);
n=0.3*randn(size(t));
x=s+n;
figure, plot(x), grid on
title('Signal with Noise')
figure, periodogram(x,[],[],fs)
title('PSD using Periodogram')
figure
X=abs(fft(x));
f=(0:length(X)-1)*fs/length(X);
plot(f(1:500),X(1:500)), grid on
title('Magnitude Spectrum')
figure
bar([mean(s.^2) var(n)])
set(gca,'XTickLabel',{'Signal','Noise'})
title('Power Comparison')
grid on
```

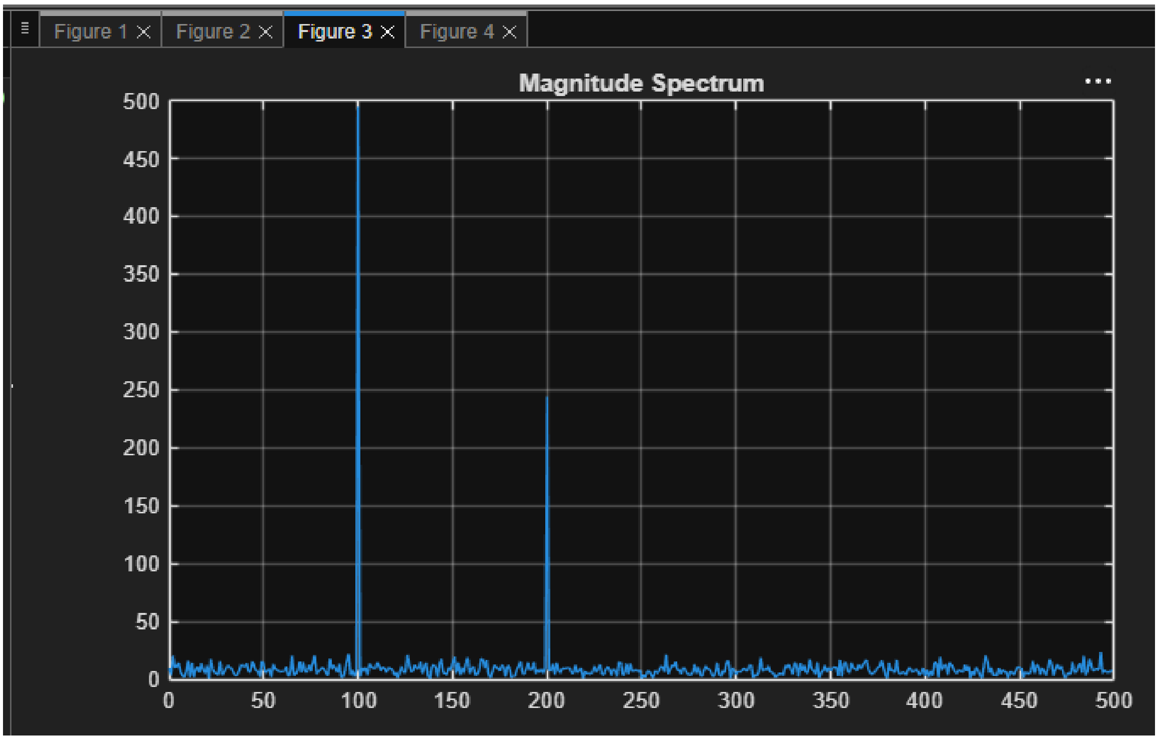
Signal with Noise



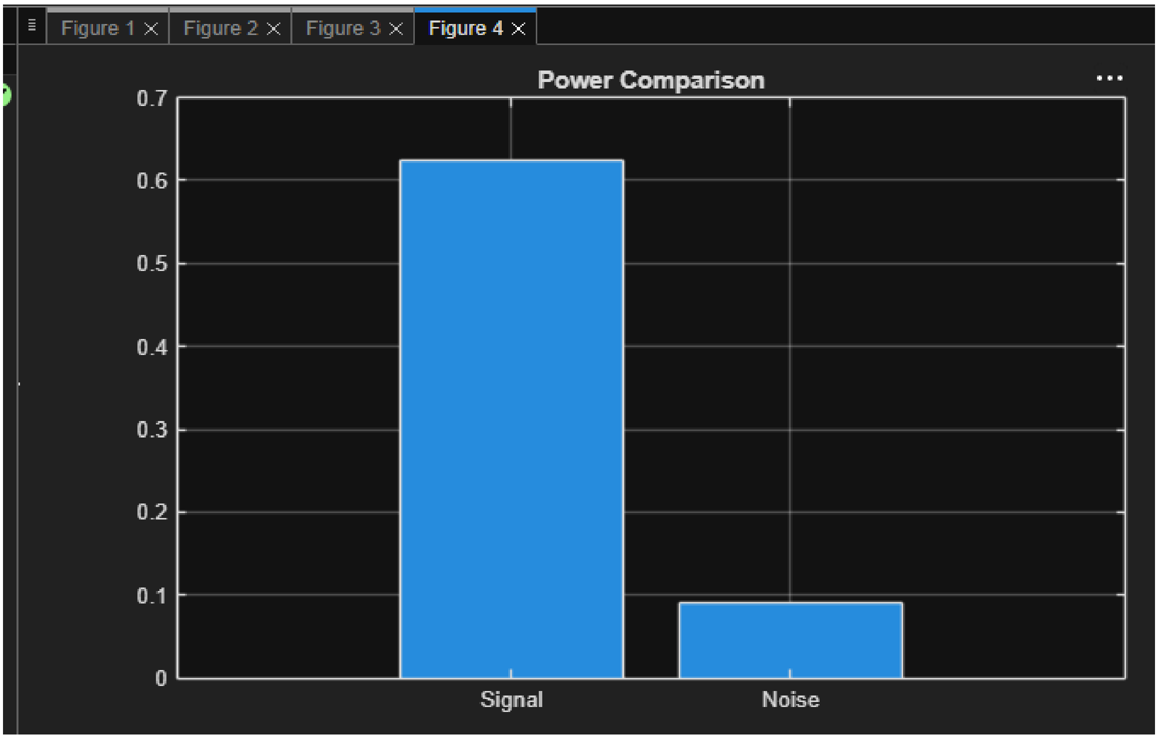
PSD using Periodogram



Magnitude Spectrum



Power Comparison



Result: The power spectral density was estimated successfully using the periodogram method. The dominant frequency components and signal/noise power were observed.